Engineering our way out of tool fatigue

Peter Gigengack



CYBERCON 2026
AUSTRALIAN CYBER CONFERENCE

Presented by AISA

Supported by VICTORIA State Government

This remains a common scene for security teams…



CYBERCON 2026
AUSTRALIAN CYBER CONFERENCE

Presented by AISA

Supported by VICTORIA State Government

Yet we've never had more security tooling…





whoami

- Seasoned Cyber & ICT professional 20+ years
- Recovering Sysadmin
- Continual learner
- Mentor

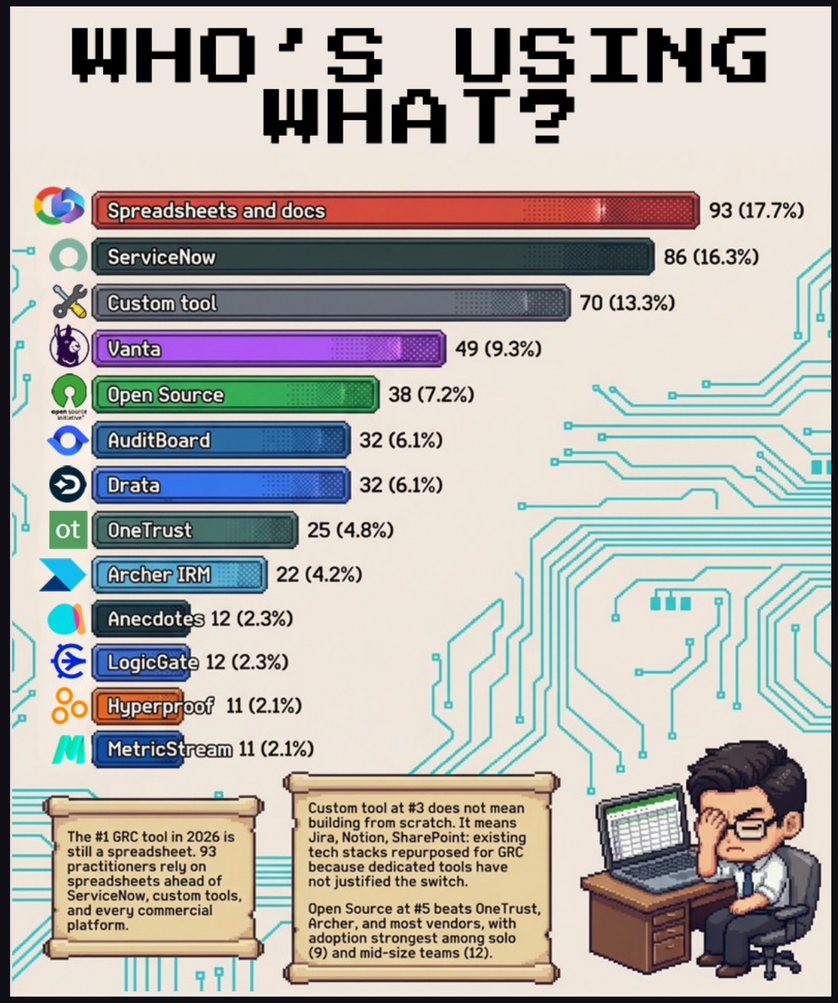
- Passionate about automation and innovation

So, if you had to answer a simple question?

Are all your crown jewels monitored?

Tool A	Yes
Tool B	Maybe
Tool C	Different Number
Spreadsheet	Last updated 2019

Yet spreadsheets remain #1 in GRC



State of GRC 2026 Report: Spreadsheets are still #1

Two separate worlds…

Governance Risk and Compliance (GRC) Teams
Policies
Risk Registers
Controls
Audits

Manual Effort

Engineering Teams
Systems
Logs
Data
APIs

What is GRC Engineering

Dimension	Traditional GRC	GRC Engineering
Primary Output	Policies, reports, spreadsheets	Automated controls, infrastructure code, dashboards
Compliance Approach	Point-in-time audits	Continuous monitoring and automated remediation
Tools	GRC platforms, Excel, Word	Terraform, Python, AWS Config, CI/CD pipelines
Evidence Collection	Manual screenshots, email chains	Automated evidence pipelines, API integrations
Scalability	Linear - more work requires more people	Exponential - automation scales across the organisation
Audit Readiness	Weeks of preparation before audits	Always audit-ready by design
Key Skill	Policy interpretation and communication	Systems thinking and automation

GRC Engineering vs Traditional GRC



Coming back to this question….

Are all your crown jewels monitored?

I started exploring GRC engineering approaches…

But I needed a problem to solve.

The Problem: Tackling SOC/SIEM Coverage

I had soooooo many questions:

- What is good visibility of Crown Jewels in Hybrid Enterprise Environments?
- Are we collecting logs (and the right logs) to identify compromises?
- Is the SIEM collecting in a timely manner?
- How do we identify and fix gaps?

Oh – And I don’t work in security operations.

I'M GONNA HAVE TO



SCIENCE THE S**T OUT OF THIS

imgflip.com

How to measure and assess?



Detecting and Mitigating Active Directory Compromises
ASD Guidance

I knew I had to start with

1. Framework
2. Best Practice Guidance
3. Combined view of assets
4. Real-time telemetry

I began designing….A simple schema

Name	Type
ASSETGROUP	Text
Description	Text
Priority	Integer

AssetsGroup

To enable categorisation of both Active Directory and Non-Active Directory Assets. It needed to work across common AD/Entra Asset Classes.

Name	Type
ASSETGROUP	Text
Description	Text
EventID	Integer

EventIDs

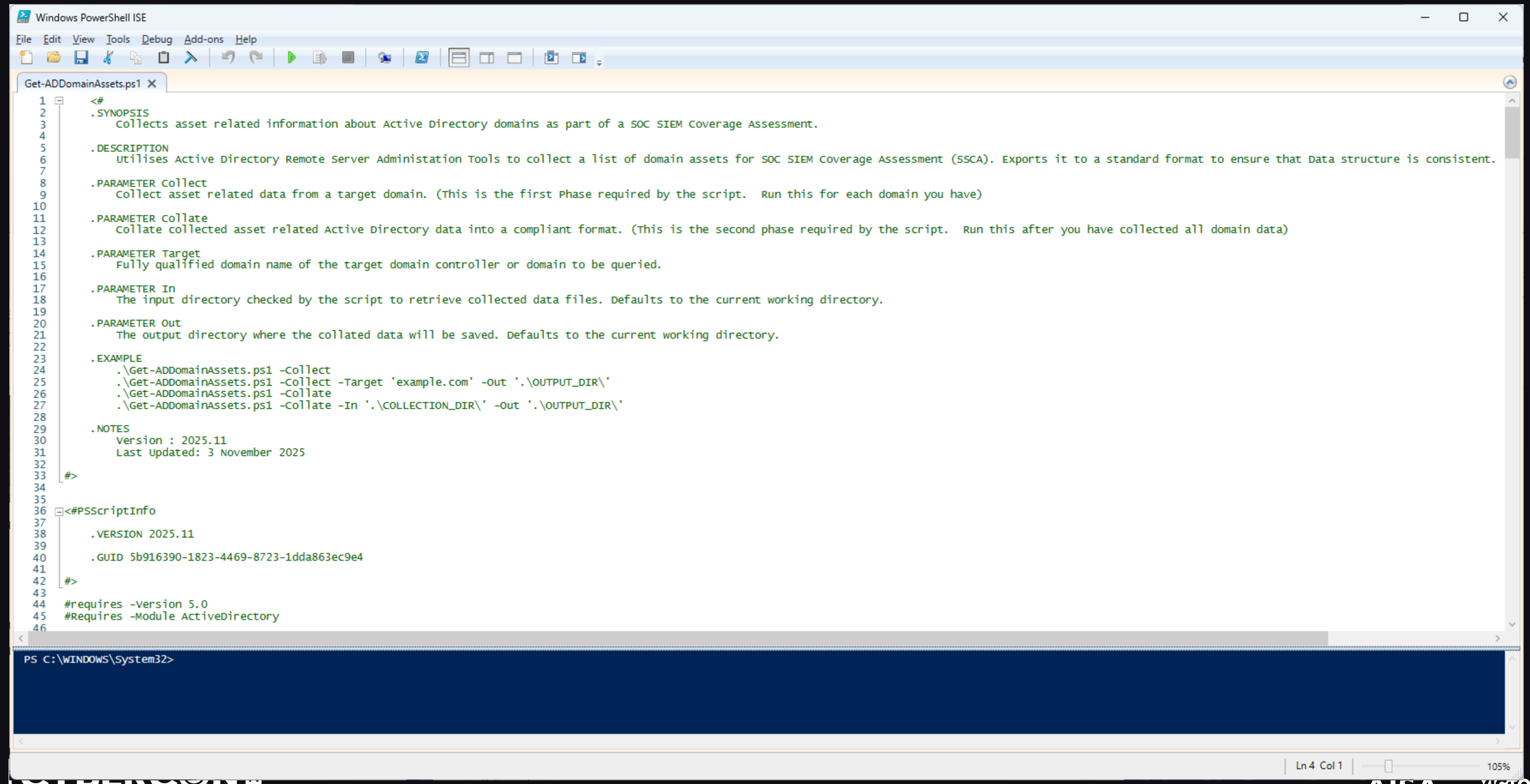
Windows EventIDs and event descriptions mapped to asset groups based on Detecting and mitigating Active Directory compromises guidance

Name	Type
ASSETGROUP	Text
AD Compromise	Text
EventID	Integer

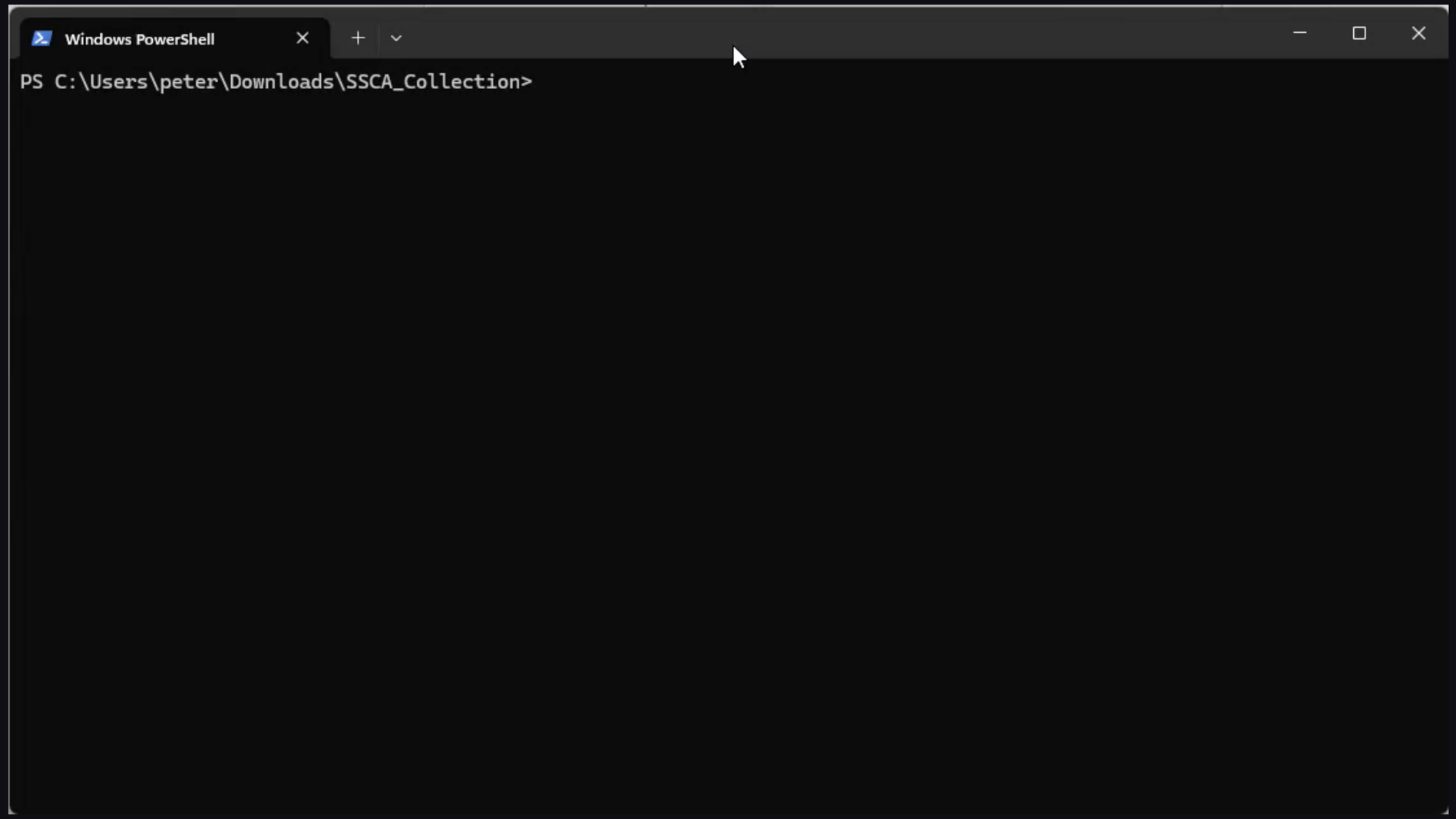
CompromiseMap

Windows EventIDs mapped to asset groups and compromise types against Detecting and mitigating Active Directory compromises guidance

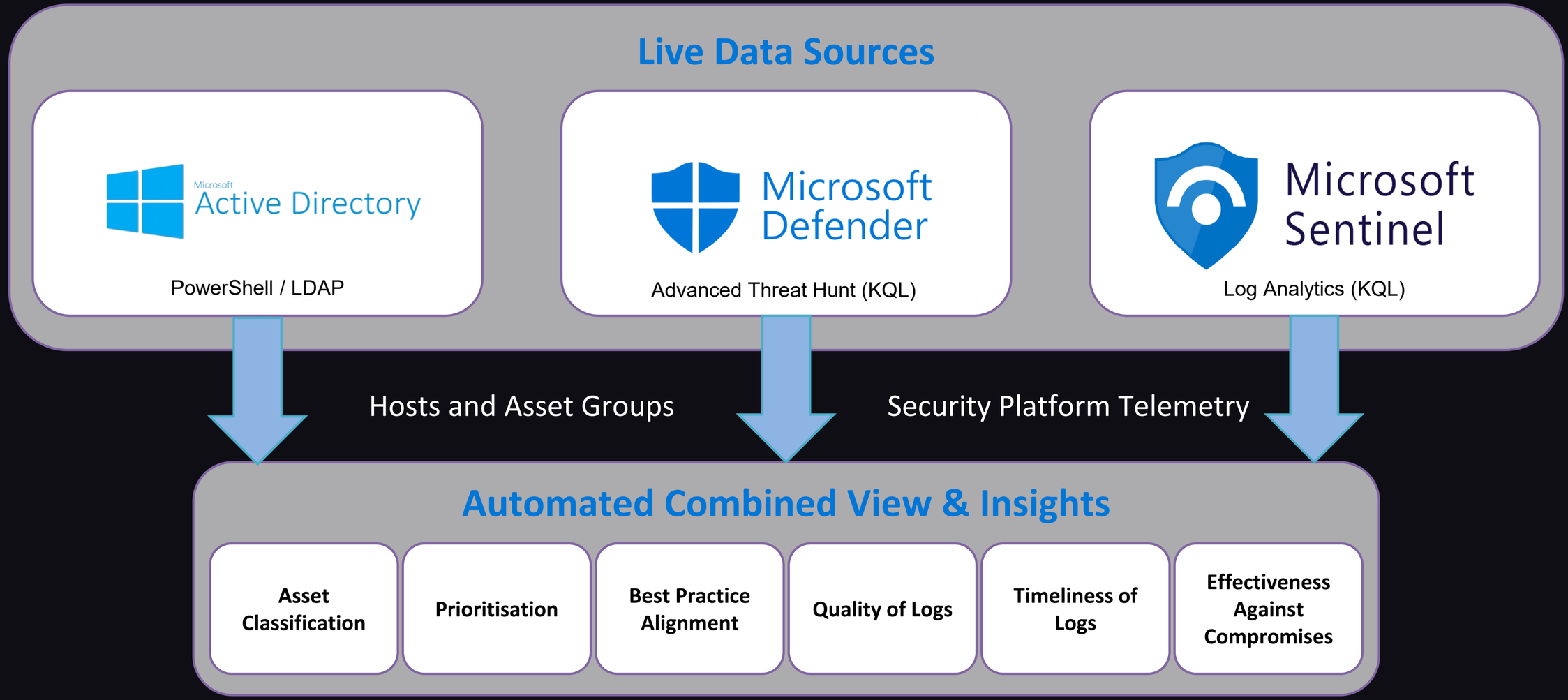
I began developing….a collection script



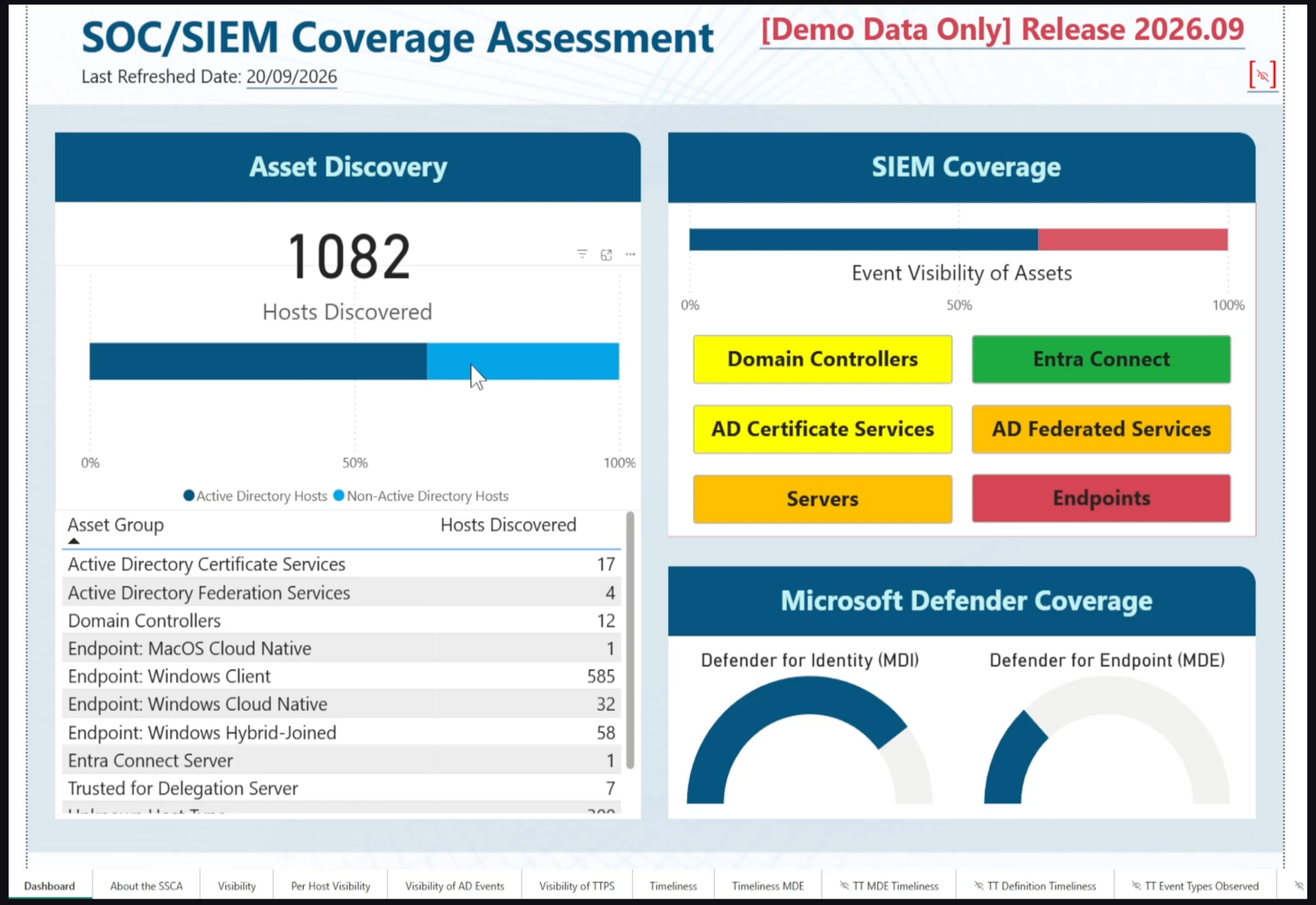
Active Directory Data Collection (Offline)



The engineering approach…



The PowerBI Dashboard



Driving better outcomes….faster

The tool developed ticked several boxes

- ✅ Had to be functional
- ✅ Effectively Identified Gaps
- ✅ Develop Remediation Actions
- ✅ Provide a path for Continuous Improvement and allow iteration

Since the initial release in 2025

- ✅ Expanded to allow for from Live Active Directory Data
- ✅ Improved documentation
- ✅ Address minor bugs in logic and collection queries
- ✅ Updated to reflect Sept 2026 Guidance, expanding detection capabilities

Driving better outcomes….faster

Before	After
Monthly Review	Continuous Assurance
Stale Spreadsheets	Real-time Metrics
Manual Validation	Automated Validation
Audit Sprints	Continuous Coverage

Since discovering GRC engineering…

It’s provided structured approaches to

- Standardising Data Collection
- Analysis and applying logic
- Creating, defining and monitoring security baselines
- Accelerating report creation

Greatest impact – Free time to work on high value tasks

Where to start

GRC Engineering Community is full of builders

GRC Playground

- Hands-on labs, a control-by-control playbook that automates the requirement end to end, and the workbook they came from.

GRC Engineering Club

- Not just a club. A movement. From checkbox GRC to engineered systems.
- Includes labs, courses, certifications and learning pathways

GRC Engineering Learning Hub

- Community-developed knowledge base of learning and career development resources for practitioners interested in advancing their GRC engineering skills and competencies.



Questions?

Connect via LinkedIn

linkedin.com/in/peter-gigengack

GitHub Resources

github.com/peteGGG/CyberCon2026